Aluminum Automatic Continuous Cutting Saw

USER MANUAL

WTMZ-450 · WTMZ-500 · English Edition 2



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Warranty

- This is the user manual for the **Aluminum Automatic Continuous Cutting Saw WTMZ-450 · WTMZ-500** supplied by CHANGHO INC.
- The warranty period is **one year** from the date of delivery. Consumables (saw blade · belt · filter) are not covered. Damage caused by user error or modification is not covered.
- For warranty repair, we will advise you on the repair method after checking the symptoms. Repair at our workshop is available only for **small machines that are easy to move**.
- Specifications may change without notice for performance improvement. If the actual machine differs from this manual, **the actual machine takes precedence**. Please let us know through the contacts in Chapter 9 and we will correct it in the next edition.

Do not use the machine without reading Chapter 1, Safety Precautions. Failure to follow them may result in serious injury. Keep this manual with the machine.

Installation, adjustment, operation, maintenance and troubleshooting are taken from the manufacturer's English manual; the source is marked [Orig. section no.]. Specifications follow CHANGHO INC product data; the control panel and parts follow photos of the actual machine and the wiring diagram.

1. Safety Precautions

1-1. Work Area

Keep the work area tidy

Chips and cutting oil on the floor are slippery. Clean up after every job.

Install indoors in a dry place

Install indoors, free of moisture and corrosive gas, at 15–35 °C, on a solid concrete floor. [Orig. 4.1]

Keep unauthorized persons away

Do not let children or people not involved in the work come near the machine.

1-2. Clothing and Protective Equipment

Wear safety glasses

Chips and cutting-oil mist can fly. Cutting is loud, so wear ear protection as well.

Avoid loose clothing, jewelry and cotton gloves

They can be caught by the rotating blade or belt. Tie up long hair.

Wear safety shoes

You will be carrying material and working on a floor covered with chips.

1-3. Machine Handling

Never run the machine with the blade cover open

Make sure the transparent cover is closed before working.

Keep hands away from the rotating blade

To adjust the material or clear chips, stop the motor and wait until the blade has **stopped completely**.

Learn where the emergency stop is first

It is the red STOP button on the right of the control panel. Press it immediately if you notice unusual noise, vibration or smoke.

Do not feed material beyond the cutting capacity

Use only within the maximum cutting size in the specifications (Chapter 3).

Cut only aluminum and PVC profiles

Steel cannot be cut. The manufacturer's manual also specifies aluminum and PVC profiles. [Orig. 1]

1. Safety Precautions (continued)

1-4. Electrical Safety

Always ground the machine

The power supply is 380 V, 3-phase. Connect according to the wiring diagram and connect the ground wire. [Orig. 4.4]

Only qualified electricians may open the control box

380 V is present inside the control box. Turn the power switch OFF before opening it.

Turn the power switch OFF before servicing

The power switch is the red handle on the yellow plate on the side of the machine.

1-5. Inspection and Servicing

Disconnect both power and air before servicing

Replace the blade, inspect or clean only after turning the power switch OFF and closing the air valve.

Do not disassemble the air service unit (FRL) while under pressure

This can cause injury or damage to the machine. [Orig. 8.3 Note]

Use only parts of the same specification

Order parts such as the saw blade and belt using the markings in the parts list (Chapter 7).

> This chapter is based on the safety standards CHANGHO INC applies to cutting saws of the same type. It cannot cover every situation, so always work with care.

2. Product Overview · Models

This machine is **an automatic saw that cuts short pieces, such as corner connectors for aluminum windows and doors, continuously to the same length**. **Aluminum and other non-ferrous metals, as well as PVC, nylon and MC nylon** can be cut. [Orig. 1 · confirmed by CHANGHO INC]

Features

1. Material clamping, material feeding and blade advance / return are all driven by **compressed air (pneumatics)**. [Orig. 3.1 · 3.3]
2. The machine uses a **carbide-tipped saw blade**. Moving parts run on linear guides, so motion is smooth and accurate. [Orig. 1]
3. The frame is welded from steel angles and steel plate. [Orig. 3.2]
4. A controller (PLC) in the electrical control box manages all operations. [Orig. 3.4]
5. You can set the number of pieces on the counter. [Control panel · wiring diagram]
6. A cutting-oil spray unit is fitted. [Catalog]

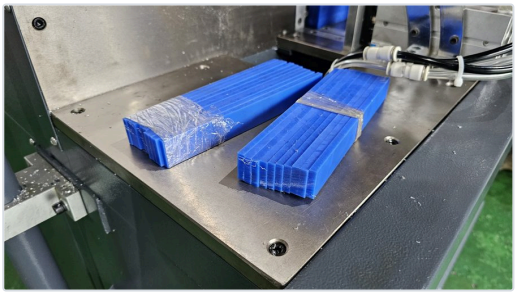
Models

Model	Saw blade	Power	Max. cut (H × W)
WTMZ-450	Φ450	3.0 kW	110 × 200 mm
WTMZ-500	Φ500	3.0 kW	130 × 200 mm

Both models share the same machine body; only the blade size (Φ450 · Φ500) differs. Maximum cutting sizes are listed in the specifications (Chapter 3).



Aluminum Automatic Continuous Cutting Saw — common appearance of WTMZ-450 · WTMZ-500



MC nylon strips cut continuously to the same length



Aluminum round tubes cut continuously

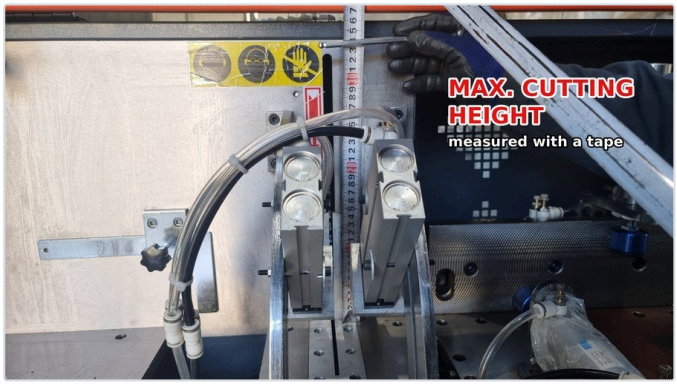
3. Specifications

Item	WTMZ-450	WTMZ-500
Power supply	380 V, 3-phase, 60 Hz	380 V, 3-phase, 60 Hz
Power	3.0 kW	3.0 kW
Motor speed	2,800 r/min	2,800 r/min
Air pressure	0.5 ~ 0.8 MPa	0.5 ~ 0.8 MPa
Saw blade	450 × 4.4/3.6 × 30 · 100 teeth (100T)	500 × 4.4/3.6 × 30 · 100 teeth (100T)
Max. cut (H × W)	110 × 200 mm	130 × 200 mm
Cutting length	Unlimited	Unlimited [Orig. 2]
Material feed speed	0 ~ 3 m/min	0 ~ 3 m/min [Orig. 2]
Machine size (W × D × H)	1500 × 1400 × 1900 mm	1500 × 1400 × 1900 mm
Weight	Approx. 500 kg	Approx. 500 kg [Orig. 2]

Blade specification: tooth thickness 4.4 / body thickness 3.6, bore 30 mm; 100T means 100 teeth. The two models differ only in blade diameter (450 · 500), and the maximum cutting height differs accordingly.



Maximum cutting width — measured with a tape



Maximum cutting height — measured with the material standing



Saw blade and cover



Blade marking — 500×4.4/3.6×30×100T

4. Installation

① Location

Install indoors, free of moisture and corrosive gas, at 15–35 °C, on a solid concrete floor. [Orig. 4.1]

② Unpacking · inspection

Move the machine to the installation site, unpack it and check the whole machine for damage before installing. [Orig. 4.2]

③ Leveling

Level the stand legs using the top surface of the worktable as reference. [Orig. 4.3]

④ Power connection

Connect 380 V, 3-phase, 60 Hz power according to the wiring diagram and always connect the ground wire. [Orig. 4.4]

⑤ Air connection

Connect 0.5–0.8 MPa air to the air inlet (air service unit) on the machine, turn on the air and check the whole machine for leaks. [Orig. 4.5]

> When power is first turned on, check the indicator lamps on the control panel. If the “Protection directive” lamp lights, do not operate; check the power connection again. A 3-phase protection relay is installed in the control box. [Control box photo · wiring diagram]

The material support table is standard equipment. The overall length of the saw plus the support table is **5,220mm** (measured on the machine shown). Plan the installation space as this length plus room for loading and unloading material.



Saw and material support table — overall length 5,220 mm



Power switch — red handle on the yellow plate on the side of the machine



Air connection — air service unit (filter · pressure regulator · lubricator)

5. Control Panel · Adjustment · Operation

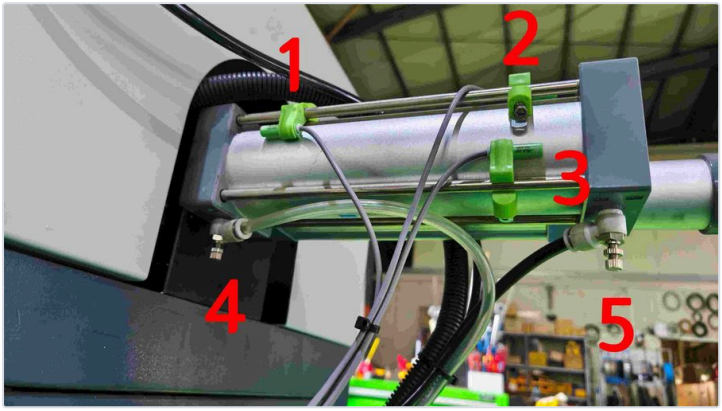


Control panel — top row: 4 lamps · counter / bottom row: 3 buttons · selector switch · emergency stop

Position	Panel label	Meaning	Circuit symbol
Top 1	Power Indicator	Power lamp	LA0
Top 2	Electric Instructions	Motor run lamp	LA1
Top 3	Automatic Instructions	Automatic operation lamp	LA2
Top 4	Protection directive	Protection lamp	LA3
Top right	COUNTER (PV · SV)	Counter — PV (current count) · SV (set count)	Counter
Bottom 1	Motor stop	Motor start / stop — toggles each time it is pressed	SB3
Bottom 2	Manual sawing	Manual cutting	SB2
Bottom 3	Click	Step operation — one step per press in Manual	SB1
Bottom 4	Manual · Auto / Job selection	Manual / Auto selector switch	SA1
Bottom 5	STOP	Emergency stop	SB0
Left of panel	—	COUNT ON/OFF switch	—

5. Control Panel · Adjustment · Operation (continued)

5-1. Sensors and Speed Control Valves



Blade feed cylinder — numbers match the descriptions on the right

① Maximum advance position
Sets the farthest forward position of the blade.

② Return sensor
Signals that the blade has returned.

③ Count signal
Sends a count signal to the counter for each cut.

④ Blade advance speed
Speed control valve — adjusts how fast the blade advances.

⑤ Blade return speed
Speed control valve — adjusts how fast the blade returns.

5-2. Adjustment

Cutting length
Set the length by moving the stop that determines where the material stops. Use the scale on the worktable and fine-tune with the adjusting knob. [Orig. 5.1]

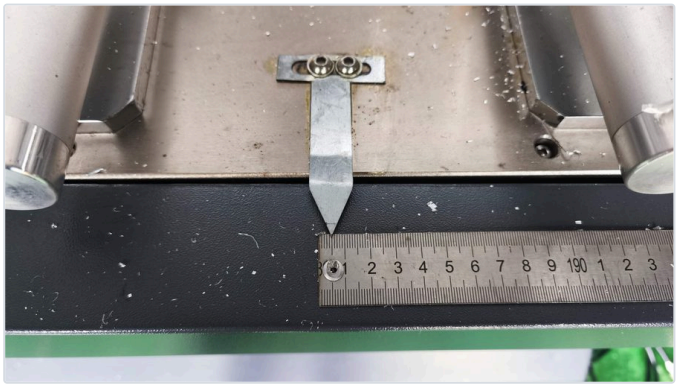
Blade feed speed
The blade is pushed by a cylinder. Adjust the speed with the screws of the speed control valves (④ · ⑤ in the photo above). **Use a faster speed for thin-walled material and a slower speed for thick-walled material.** [Orig. 5.2]

Cutting-oil spray amount
The blade must receive the right amount of cutting oil. Adjust the spray amount at the nozzle. **Use only neat (non-water-soluble) cutting oil.** [Orig. 5.3]

Other parts
These are adjusted before shipment and need no adjustment after installation. If adjustment is needed, please contact us. [Orig. 5.4]



Cutting length scale and position stop



Scale at the material stop position

5. Control Panel · Adjustment · Operation (continued)

5-3. Preparation

① Check before starting

Check that the air compressor pressure is **at least 0.5 MPa**. If it is lower, raise the compressor pressure first. Check the oil level at each lubrication point and in the cutting-oil tank. [Orig. 6]

② Power on

Turn the main power switch on the side of the machine to ON. The “Power Indicator” lamp lights.

③ Set length and quantity

Set the cutting length with the position stop and enter the number of pieces in the counter SV. To count pieces, set the COUNT switch on the left of the control panel to ON. When the set number has been cut, the machine returns to its starting state and stops.

5-4. Manual Step Operation — “Click” button

Set the selector switch to **Manual** and press the **“Click” button**. Each press moves the machine one step in the order below. Use this method to check each motion one by one at first installation or when setting the length.

1. Motor start
2. Material clamp
3. Move the gripped material by the set length
4. Clamp at the cutting position (blade-side clamp)
5. Blade advance · return — cut
6. Material feed unit returns
7. Blade-side clamp releases
8. Repeat from the first step



Material clamping unit

5. Control Panel · Adjustment · Operation (continued)

5-5. Automatic Continuous Cutting

① Main power ON

Turn on the main power switch on the side of the machine.

② Selector switch to Auto

Turn the selector switch to Auto.

③ Check emergency stop is released

Make sure the red STOP button is not pressed in. If it is, turn it to release it.

④ Check material position

Make sure the material is in place on the support table.

⑤ Motor start

Press the “Motor stop” button at the lower left of the control panel. The motor starts and **automatic continuous cutting begins**.

To stop, press the same “Motor stop” button once more. In an emergency, press the red STOP button.

> Despite its name, the “Motor stop” button on the control panel **both starts and stops the motor**. Do not put your hands near the blade or the material feed unit during automatic operation.



Control panel — bottom left is motor start / stop (Motor stop), third is “Click”



COUNT ON/OFF switch — left side of the control panel



Material clamping unit — seen from above

5. Control Panel · Adjustment · Operation (continued)

5-6. Material Sensor — Automatic Stop When Material Runs Out

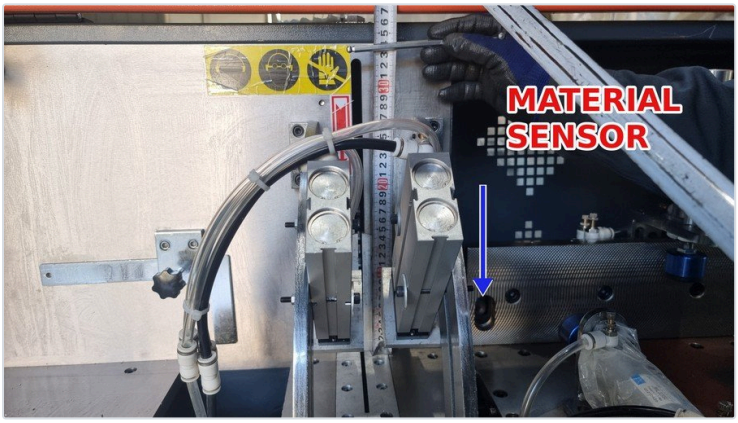
The material sensor checks whether there is material left to cut (blue arrow in the photo below). When the end of the material slides past the sensor during automatic continuous cutting, **the sensor turns OFF, the machine judges that there is no material to cut, and the saw stops.**

① Automatic stop
When the sensor turns OFF, the saw stops. This is not a fault.

② Align material
Place new material on the support table and align it again.

③ Restart
Restart the saw. The start procedure is the same as 5-5 ⑤.

> After aligning the material, **remove your hands from the blade and the material feed unit before restarting.**



Material sensor — where the blue arrow points

If the Material Sensor Does Not Work — Check Procedure

If the saw stops even though material is present, or does not stop when the material has run out, check in the order below. **Check with the motor stopped and the power on.**

① Sensor indicator lamp
Place material in front of the sensor and remove it, and watch whether the small lamp at the cable end of the sensor turns on and off.

② Controller X0 input lamp
Watch whether the X0 lamp on the controller (PLC) inside the control box turns on and off at the same time. The material sensor is SQ0 in the wiring diagram and is connected to controller input X0 (Chapter 7, wiring diagram).

③ Clean the sensor face
If there is cutting oil or aluminum dust on the sensor face, wipe it off.

> If the sensor lamp never lights, the sensor or its wiring is the likely cause. Sensor specifications and the replacement procedure are on pages 13–14.

> 380 V is present inside the control box. **Check ② must be done by a qualified electrician** (see 1-4).

5. Control Panel · Adjustment · Operation (continued)

5-6. Material Sensor — Specifications and Replacement

The material sensor is the **AE3F-DS10C4**, a photoelectric sensor that detects material by light reflected back to its lens. A dirty lens can stop it from detecting material, so keep the lens clean.



Material sensor AE3F-DS10C4



Sensor label — model · voltage · sensing distance



Sensor label — wire colors (brown · black · blue)

Item	Specification
Model	AE3F-DS10C4 (TAHUA)
Type	M18 cylindrical, diffuse reflective photoelectric sensor
Output	NPN, normally open (NO) — ON when material is present
Sensing distance	10 cm (100 mm)
Supply voltage	6–36 V DC (24 V DC on this machine)
Max. current	200 mA
Wiring diagram	SQ0 · controller input X0

Wire color	Connection
Brown	DC + (24 V)
Blue	0 V
Black	Signal output → controller X0

Sensor wiring must be done **by a qualified electrician with the power off**.

5. Control Panel · Adjustment · Operation (continued)

Replacement Sensors from Other Makers

Product	Specifications · setting
Autonics BRQM100-DDTA (recommended)	Diffuse 100 mm · NPN · 10–30 V DC · set Light ON with the white wire
Autonics BR100-DDT	Diffuse 100 mm · NPN · 12–24 V DC · set Light ON with the white wire · earlier model
Panasonic CY-121A	Diffuse 100 mm · NPN Light ON · 12–24 V DC · 3 wires, connect as is · discontinued, check stock
Omron E3FA-DN11 2M	Diffuse 100 mm · NPN · 10–30 V DC · set Light ON with the mode wire
SICK VTE18-4N4212	Diffuse · NPN · 10–30 V DC · adjustable range (max. approx. 400 mm), reduce to suit · set Light ON with the white wire

When choosing: **M18 cylindrical · diffuse reflective · NPN · Light ON (normally open) · approx. 100 mm range · works at 24 V DC**. PNP versions (model name ending in -P) cannot be used. For 4-wire sensors, check the setting of the extra wire in that sensor's instructions.

Sensor Replacement Procedure

① **Power off**	Turn the power switch OFF.
② **Record the wiring**	Photograph where the three wires are connected, then disconnect them.
③ **Replace the sensor**	Loosen the two M18 nuts and fit the new sensor at the same position and depth.
④ **Connect the wires**	Connect wires of the same color, following the wire color table on page 13.
⑤ **Test**	Power on, then place and remove material in front of the sensor: the lamp must turn on and off. If there is a sensitivity adjuster, set it so the lamp lights only when material is present.

Sensor wiring must be done **by a qualified electrician with the power off**.

If you need a replacement sensor, please contact us (Chapter 9).

5. Control Panel · Adjustment · Operation (continued)

5-7. Using the Counter — Setting the Number of Pieces

Enter the number of pieces to cut in the counter (CLD-C01) at the upper right of the control panel. The counter counts every piece cut, and **when the set number has been cut, the machine returns to its starting state and stops.** To count pieces, set the COUNT ON/OFF switch on the left of the control panel to ON. From the left, the buttons are C · ∧ · > · SET.

Display · button	Function
Present value PV (top)	Number of pieces cut so far
Set value SV (bottom)	Number of pieces to cut
C button (1st)	Resets PV to 0
∧ button (2nd)	Increases the blinking digit
> button (3rd)	Moves to the next digit
SET button (4th)	Starts setting; press again to save



Counter CLD-C01 — top PV · bottom SV · 4 buttons

Setting the count

① COUNT ON/OFF switch ON

Set the COUNT ON/OFF switch on the left of the control panel to ON.

② Clear the count

Press the C button to reset PV to 0.

③ Start setting

Press the SET button; the digit to be changed in SV starts blinking.

④ Change the number

Use the > button to move to the digit to change, and the **∧ button** to increase it. Pressing once more at 9 returns it to 0.

⑤ Save

When the desired number is shown, press the **SET button** to save.



② PV reset to 0 with the C button



③ Press SET to start setting



④ Changing the number — the digit being changed blinks



⑤ Saved with SV 1350

> Every time you start a new job, reset PV to 0 with step ② (C button) before starting.

6. Pneumatics · Maintenance

6-1. Air Service Unit (FRL)

This machine uses an **air service unit (FRL)** — filter · pressure regulator · lubricator — to clean the air and regulate the pressure. Compressed air contains water, oil and dust; without filtering, the life of pneumatic parts is greatly reduced. [Orig. 7]

Filter

Removes dust and moisture (filtration 50 μm or finer). A drain valve is under the water bowl. [Orig. 7]

Pressure regulator

Reduces the pressure to the required level and keeps it constant. [Orig. 7]

Lubricator

Sprays oil as a mist and sends it with the air to the valves and cylinders. [Orig. 7]

6-2. Using the Air Service Unit

- Install the air service unit upright **with the drain valve at the bottom**. Air must flow only from P (inlet) to A (outlet); do not install it in reverse. [Orig. 8.3]
- Press the drain valve under the water bowl (marked F in the original) to **drain water frequently**. If no water comes out, the valve is blocked; remove the bowl and clean it. [Orig. 8.3]
- To change the pressure, pull the knob up, then **turn it toward "+" to raise the pressure and toward "−" to lower it.** When the gauge shows the desired pressure, push the knob down to lock it. If the pressure cannot be set, first check that the supply pressure is normal before judging the valve faulty. [Orig. 8.3]
- For the lubricator, **4–10 drops per minute** is sufficient. Keep the oil level between the upper and lower marks on the bowl, and clean the bowl **once a week**. [Orig. 8.3]

> Disassembling the air service unit under pressure can cause injury or damage the machine. Always shut off the air and release the remaining air before disassembly. [Orig. 8.3 Note]

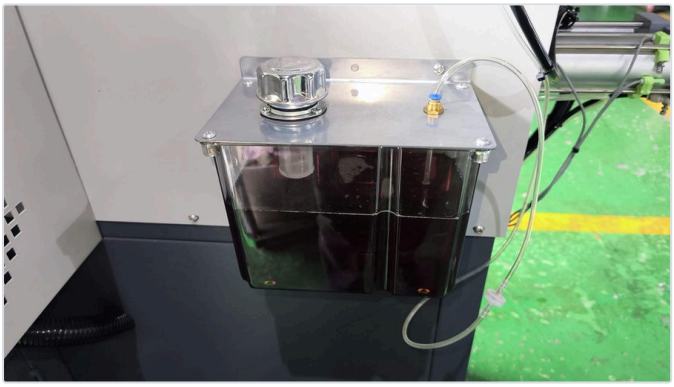
6. Pneumatics · Maintenance (continued)

6-3. Inspection Schedule

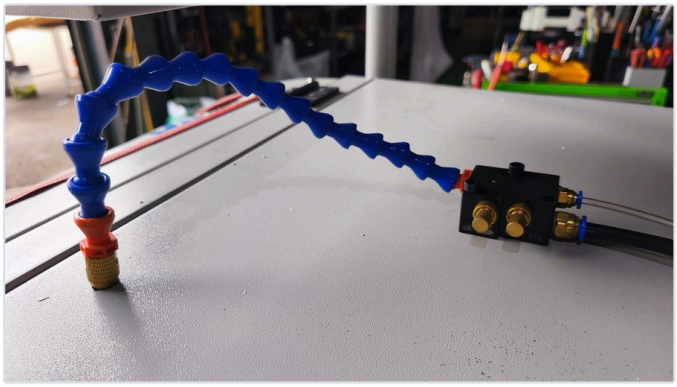
Interval	Check item	Action
Daily, before work	Lubrication points · lubricator oil · air pressure · supply voltage	Refill if low; do not operate if pressure or voltage is incorrect [Orig. 8.2]
As needed	Water bowl of air service unit	Drain the water [Orig. 8.3]
Once a week	Lubricator oil bowl	Clean [Orig. 8.3]
Every 15 days of operation	Main fixing screws	Check all for looseness [Orig. 8.1]
Regularly	Filter · water bowl	Clean and drain [Orig. 8.4]
Before long idle periods	Whole machine · guide rods	Clean, then apply rust protection to guide rods and moving parts [Orig. 8.5]
When cutting oil runs low	Tank oil level	Refill with neat cutting oil

6-4. Cutting Oil

Use only **neat cutting oil (not mixed with water)**. Check the oil level in the tank and refill; adjust the spray amount at the nozzle.



Cutting-oil tank



Cutting-oil spray unit — nozzle and control valve

6. Pneumatics · Maintenance (continued)

6-5. Blade Replacement

① Disconnect power · air
Turn the power switch OFF and close the air valve. Make sure the blade has stopped completely.

② Check blade direction
Open the cover and first check the rotation arrow marked on the blade.

③ Loosen the nut
While holding the blade so it cannot turn, loosen the fixing nut by turning it **in the direction of blade rotation**.

④ Fit the new blade
Fit the blade with its rotation arrow in the correct direction and tighten with a tool only enough that it does not come loose. The blade tightens itself as it rotates.

⑤ Test run
Close the cover, reconnect power and air, then run it once manually to check for vibration and noise.



Blade fixing nut and rotation arrow

Do not loosen the nut with a large automotive impact wrench. If you force it in the wrong direction, the thread will be damaged beyond repair.

The blade replacement procedure is based on the standard CHANGHO INC applies to circular saws of the same type.

7. Structure and Parts — Parts List · Consumables

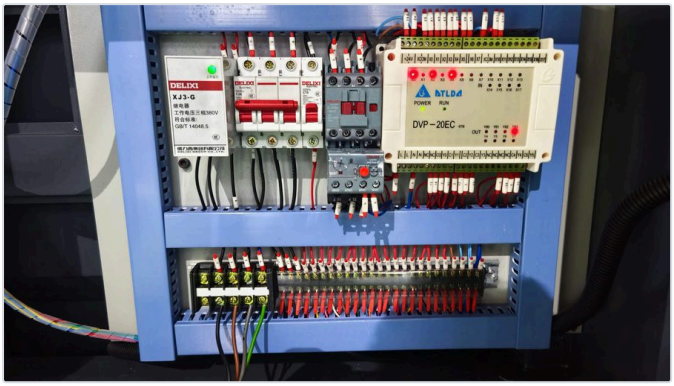
No.	Part name	Original marking	Qty
1	Saw blade	WTMZ-450: 450×4.4/3.6×30×100T WTMZ-500: 500×4.4/3.6×30×100T	1
2	Drive belt	13X620Li (675La)	2
3	Counter	CLD-C01 (AC220V · 0~9999)	1
4	Controller (PLC)	DVP-20EC	1
5	Phase protection relay	XJ3-G	1
6	Circuit breaker	QF1	1
7	Magnetic contactor	KM1	1
8	Fuse	FU0 1A · FU1 2A · FU2 2A	3
9	Blade feed solenoid valve	YV2	1
10	Material clamp solenoid valve (low pressure)	YV3	1
11	Material clamp solenoid valve (high pressure)	YV4	1
12	Material feed solenoid valve	YV5	1
13	Material position sensor	SQ0 · AE3F-DS10C4	1
14	Material feed forward limit	SQ1	1
15	Blade home limit	SQ2	1
16	Blade return limit	SQ3	1
17	Air service unit	Filter · Regulator · Lubricator	1
18	Cutting-oil spray unit · tank	—	1

No. 1 is taken from the marking on the actual blade, No. 2 from the belt marking, and Nos. 3–16 from photos of the control box and the wiring diagram legend. The WTMZ-450 blade differs from the WTMZ-500 blade only in diameter (450); thickness, bore and number of teeth are the same.

Consumables

The saw blade, drive belt and FRL filter are consumables. For the same diameter, blades for aluminum have thicker teeth and blades for PVC thinner, so order the blade that matches your material.

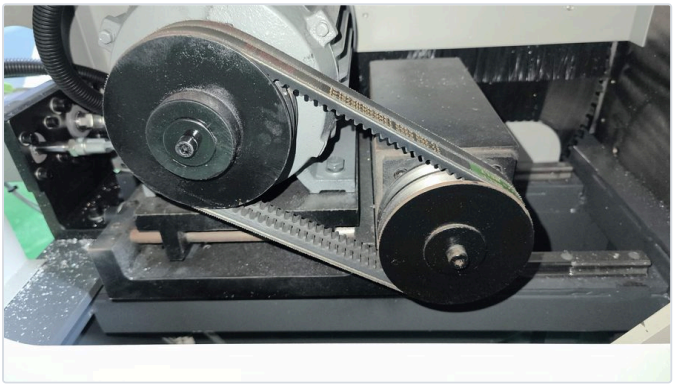
7. Structure and Parts — Control Box · Drive



Inside the electrical control box



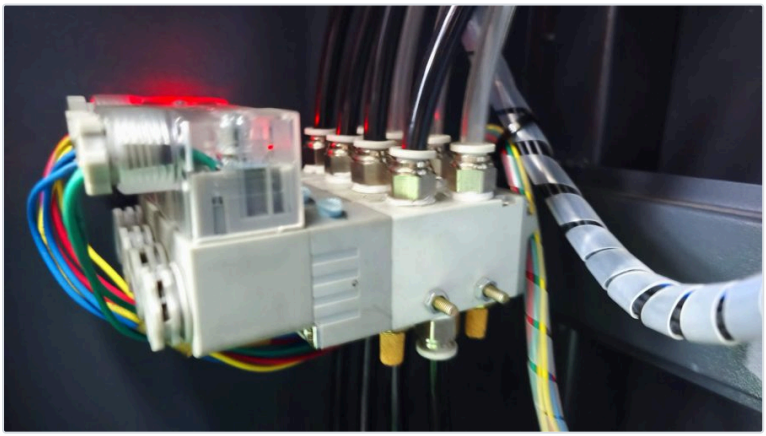
Controller (PLC), contactors · breakers



Drive belt connecting the motor and the blade shaft

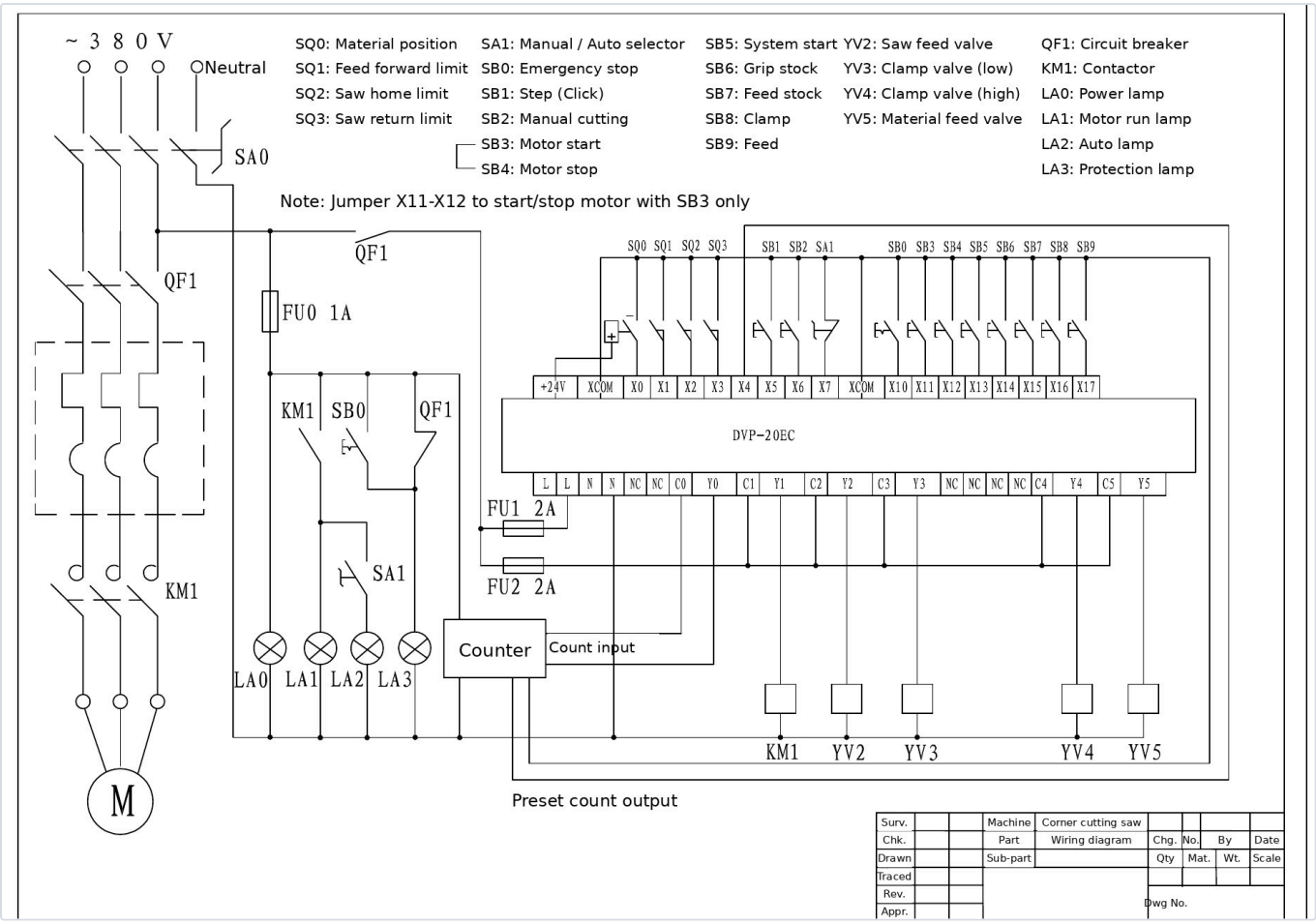


Belt marking — 13X620Li (675La)



Solenoid valve block

7. Structure and Parts — Wiring Diagram



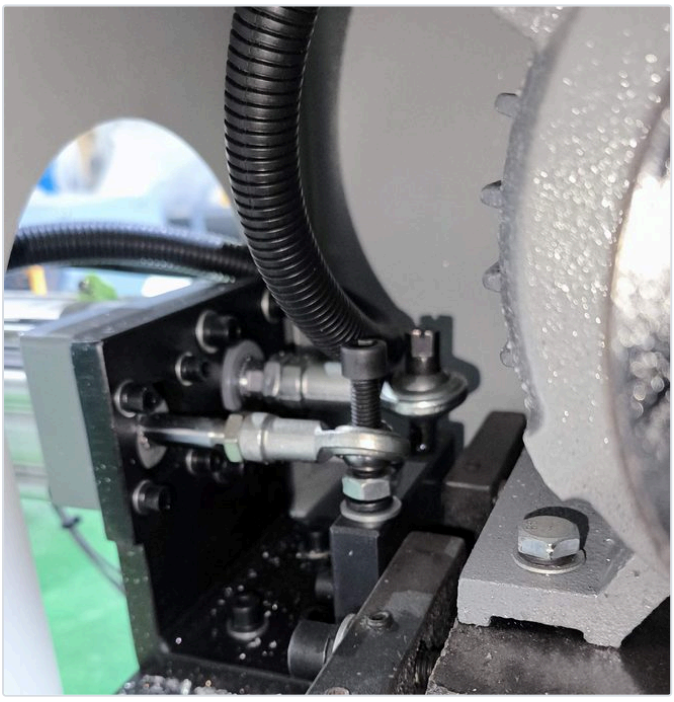
Wiring diagram — text of the manufacturer's original drawing translated into English

Symbol	Meaning	Symbol	Meaning
SQ0	Material position sensor	SB5	System start
SQ1	Material feed forward limit	SB6	Grip stock
SQ2	Blade home limit	SB7	Feed stock
SQ3	Blade return limit	SB8	Material clamp
SA1	Manual / Auto selector	SB9	Feed
SB0	Emergency stop	YV2~YV5	Solenoid valves (parts list above)
SB1	Step operation	QF1 · KM1	Circuit breaker · magnetic contactor
SB2	Manual cutting	LA0~LA3	Power · run · auto · protection lamps
SB3 · SB4	Motor start · stop	Counter	Counter (count input · preset count reached signal)

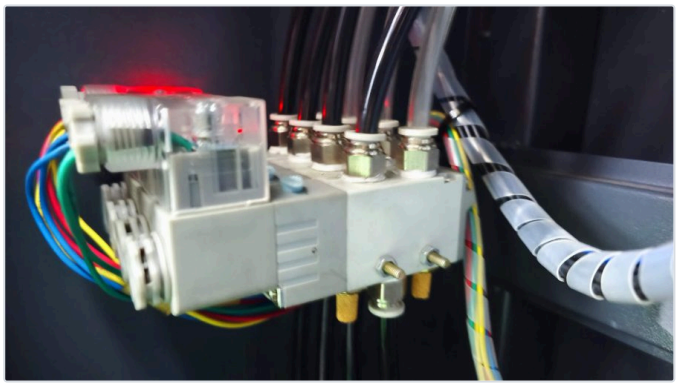
Note on the diagram: connecting X11 and X12 lets SB3 alone start and stop the motor. The Chinese text in the diagram has been translated into English, and the symbols are summarized again in the table above.

8. Troubleshooting

Symptom	Check item	Action
Pressure does not rise	Regulating spring of the pressure regulator	Replace immediately if broken [Orig. 9.1]
Solenoid valve makes noise	Between the moving and fixed iron cores in the valve	Remove trapped debris [Orig. 9.2]
Material is not fed automatically	Automatic feed unit · material feed valve (YV5)	Repair the faulty part — contact us if difficult [Orig. 9.3]
Material clamp does not release after the blade returns	① Whether the slide cylinder return sensor detects ② Material clamp solenoid valve blocked · coil burnt out	If the sensor does not detect, contact us. Find the cause of the valve fault and replace it immediately [Orig. 9.4]
“Protection directive” lamp lights and the machine does not run	Power connection	Stop operation, check the power connection and contact us
Rough cut surface or overheated blade	Cutting-oil spray amount · blade wear · feed speed	Increase the spray amount and slow the feed. Replace the blade if worn [Orig. 5.2 · 5.3]
Air leaks	Piping · fittings · air service unit	Refit the fittings and replace leaking hoses [Orig. 4.5]
Saw stops by itself during automatic operation	Whether the material has run out and its end has passed the material sensor	Not a fault. Align the material again and restart (5-6)
Saw stops although material is present, or does not stop when material runs out	Material sensor (SQ0) lamp · controller X0 lamp · dirt on sensor face	Follow the check procedure in 5-6. If the sensor lamp does not light, replace the sensor (AE3F-DS10C4)



Slide cylinder connection — check point for the material feed unit



Solenoid valve block — check point for valve noise · blockage

If the problem is not solved by the actions in the table, or if **the electrical control box or a cylinder must be disassembled**, do not repair it yourself; contact us (Chapter 9). Disassembly or modification by the user voids the warranty.

9. Open Items · Contact

9-1. Items Left Open in This Edition

In this edition, **no values have been left blank.** If you have any questions, contact us and we will check and reply.

> If new items need confirmation, we will revise this manual and send you a new edition (this is English Edition 2).

9-2. Contact

Item	Contact
Company	**CHANGHO INC**
Address	21, Sangdong-ro 375beon-gil, Sangdong-myeon, Gimhae-si, Gyeongsangnam-do, Republic of Korea
Phone	+82-55-327-4112 (main) · +82-10-4465-4112 (mobile)
Fax	+82-55-327-3112
E-mail	geonics@hanmail.net
Website	www.changhoinc.com

> When ordering parts or asking technical questions, please give us the **model name (WTMZ-450 / WTMZ-500)** and **the No. and original marking from the parts list in Chapter 7**; this helps us respond faster.